

HDOC – 12

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Subject – C Programming

❖ PROBLEM STATEMENT:

Prepare the algorithm and test case table. Then write, compile, execute, and test the C program against the prepared use cases.

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice.

- 1) Find the sum and product of the first and last digit.
- 2) Count how many digits are lesser than, equal, and greater than a given digit `k`.

➤ We must create a C program that takes a positive integer as input and:

- 1) Provide sum and product of the first and last digit of integer as output.
- 2) Provide the counts of how many digits in integer are lesser than, greater than or equal to 'k'.

○ Algorithm:

- 1) Start.
- 2) Declare variables num, product, sum, choice, temp, last_digit, first_digit, k, greater, lesser, equal as integer.
- 3) Read num.
- 4) Display the menu:
 - 4.1) 1: Find sum and product of the first and last digit.
 - 4.2) 2: Count digits less than, equal to, and greater than k.

5) Read choice.

5.1) Check if choice = 1, if it is

- Find the last digit, first digit.
- Find the sum of first and last digit.
- Find the product of first and last digit.
- Display the result, go to step 6.

5.2) else Check if choice = 2, if it is

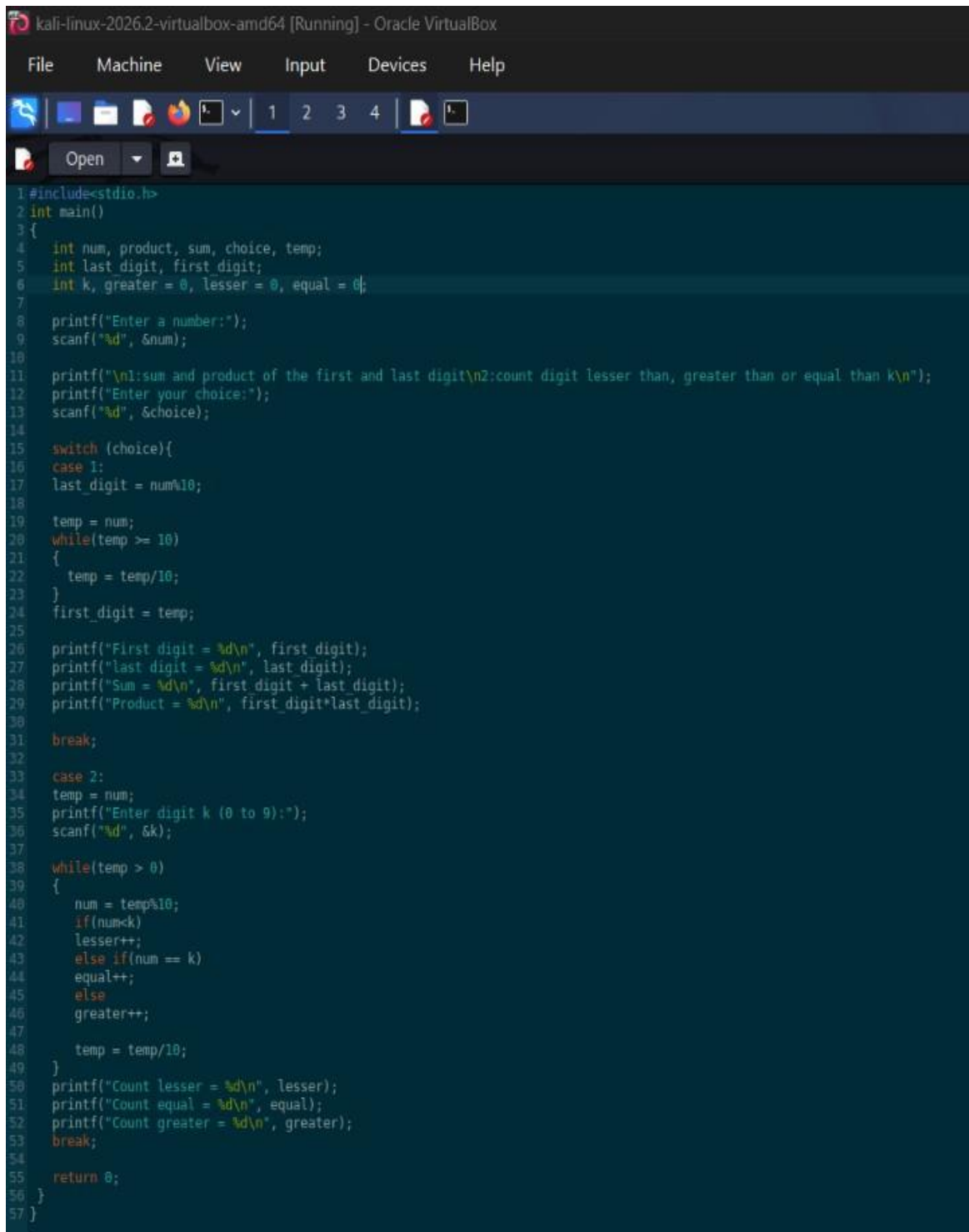
- Read k.
- Check each digit of the number.
- Count digits less than, greater than and equal to k.
- Display the counts, go to step 6.

6) Stop.

○ Test Case Table:

Test Case	Input(num)	Choice	k	Expected Output	Actual Output	Result
1.	25984	1	_____	Sum – 6 Product - 8	Sum – 6 Product - 8	Pass
2.	25984	2	4	Count_lesser – 1 Count_equal – 1 Count_greater - 3	Count_lesser – 1 Count_equal – 1 Count_greater - 3	Pass
3.	190876	1	_____	Sum – 7 Product - 6	Sum – 7 Product - 6	Pass
4.	190876	2	6	Count_lesser – 2 Count_equal – 1 Count_greater - 3	Count_lesser – 2 Count_equal – 1 Count_greater - 3	Pass

Here's is the code for this question:



```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VirtualBox
File  Machine  View  Input  Devices  Help

1 2 3 4

Open

1 #include<stdio.h>
2 int main()
3 {
4     int num, product, sum, choice, temp;
5     int last_digit, first_digit;
6     int k, greater = 0, lesser = 0, equal = 0;
7
8     printf("Enter a number:");
9     scanf("%d", &num);
10
11     printf("\n1:sum and product of the first and last digit\n2:count digit lesser than, greater than or equal than k\n");
12     printf("Enter your choice:");
13     scanf("%d", &choice);
14
15     switch (choice){
16     case 1:
17         last_digit = num%10;
18
19         temp = num;
20         while(temp >= 10)
21         {
22             temp = temp/10;
23         }
24         first_digit = temp;
25
26         printf("First digit = %d\n", first_digit);
27         printf("last digit = %d\n", last_digit);
28         printf("Sum = %d\n", first_digit + last_digit);
29         printf("Product = %d\n", first_digit*last_digit);
30
31         break;
32
33     case 2:
34         temp = num;
35         printf("Enter digit k (0 to 9):");
36         scanf("%d", &k);
37
38         while(temp > 0)
39         {
40             num = temp%10;
41             if(num<k)
42                 lesser++;
43             else if(num == k)
44                 equal++;
45             else
46                 greater++;
47
48             temp = temp/10;
49         }
50         printf("Count lesser = %d\n", lesser);
51         printf("Count equal = %d\n", equal);
52         printf("Count greater = %d\n", greater);
53         break;
54
55     return 0;
56 }
57 }
```

Compile the code using:

`gcc hdoc12.c -o hdoc12`

```
kali-linux-2026.2-virtualbox-amd64 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
kali@kali: ~/Desktop/HDOC
Session Actions Edit View Help
(kali@kali)-[~/Desktop/HDOC]
$ gedit hdoc12.c
** (gedit:44564): WARNING **: 07:58:20.286: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:44564): WARNING **: 07:58:20.286: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
(kali@kali)-[~/Desktop/HDOC]
$ gcc hdoc12.c -o hdoc12
```

Execute the code using:

`./hdoc12`

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc12
Enter a number:25984

1:sum and product of the first and last digit
2:count digit lesser than, greater than or equal than k
Enter your choice:1
First digit = 2
last digit = 4
Sum = 6
Product = 8

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc12
Enter a number:25984

1:sum and product of the first and last digit
2:count digit lesser than, greater than or equal than k
Enter your choice:2
Enter digit k (0 to 9):4
Count lesser = 1
Count equal = 1
Count greater = 3
```

```
(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc12
Enter a number:190876

1:sum and product of the first and last digit
2:count digit lesser than, greater than or equal than k
Enter your choice:1
First digit = 1
last digit = 6
Sum = 7
Product = 6

(kali@kali)-[~/Desktop/HDOC]
$ ./hdoc12
Enter a number:190876

1:sum and product of the first and last digit
2:count digit lesser than, greater than or equal than k
Enter your choice:2
Enter digit k (0 to 9):6
Count lesser = 2
Count equal = 1
Count greater = 3
```